

COLLINS BAY, SK, Canada

WMO: 710750

Lat: 58.229N

Lon: 103.678W

Elev: 413

StdP: 96.47

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.3	(c) -35.1	(d) -41.3	(e) 0.1	(f) -37.3	(g) -39.0	(h) 0.1	(i) -35.1	(j) 8.5	(k) -18.0	(l) 7.7	(m) -18.9	(n) 2.4	(o) 290	(p) 0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 25.8	(d) 15.4	(e) 24.0	(f) 14.6	(g) 22.1	(h) 14.0	(i) 17.1	(j) 22.5	(k) 16.1	(l) 21.3	(m) 15.2	(n) 20.0	(o) 3.9	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.3	18.4	14.2	10.6	17.5	13.2	9.9	16.9	49.8	22.5	46.9	21.3	44.3	20.2	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	7.0	6.3	-40.1	29.4	2.9	1.8	-42.2	30.6	-43.9	31.7	-45.5	32.7	-47.6	34.0
			-40.3	18.9	2.9	1.1	-42.3	19.7	-44.0	20.3	-45.7	21.0	-47.8	21.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-22.8	-20.1	-13.3	-3.8	4.4	12.5	15.7	14.1	8.0	-0.5	-11.6	-19.6
	DBStd	14.98	8.12	7.42	8.31	7.42	5.90	4.51	3.47	3.50	4.68	5.05	6.97	8.15
	HDD10.0	5205	1018	843	722	414	192	26	2	6	91	326	648	918
	HDD18.3	7810	1276	1077	980	664	432	181	94	138	310	583	898	1176
	CDD10.0	464	0	0	0	1	18	100	180	132	32	1	0	0
	CDD18.3	27	0	0	0	0	1	6	14	6	1	0	0	0
	CDH23.3	222	0	0	0	0	8	47	111	51	6	0	0	0
	CDH26.7	29	0	0	0	0	2	4	16	7	1	0	0	0
Wind	WSAvg	3.2	2.9	3.2	3.4	3.5	3.4	3.2	3.0	3.1	3.5	3.4	3.0	2.8
Precipitation	PrecAvg	533	20	18	27	26	41	54	82	89	66	48	30	19
	PrecMax	746	44	48	79	78	143	128	160	176	165	89	64	39
	PrecMin	362	1	2	2	3	7	8	29	18	23	20	7	0
	PrecStd	105	12	12	19	21	32	29	30	41	35	19	16	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	0.1	8.2	15.3	23.7	27.1	28.9	27.7	23.7	14.6	4.9	-0.6
		MCWB	-0.9	-0.9	3.0	7.2	12.3	15.2	16.7	16.4	14.8	8.9	2.3	-1.0
	2%	DB	-4.1	-3.0	4.4	11.2	20.0	24.6	26.1	24.7	19.7	10.8	1.1	-3.8
		MCWB	-4.4	-3.6	0.7	4.9	10.4	14.0	15.8	15.5	12.9	7.1	0.1	-4.2
	5%	DB	-7.4	-5.9	1.6	8.7	17.0	22.5	24.3	22.3	17.2	8.0	-0.5	-6.1
		MCWB	-7.7	-6.5	-0.7	3.8	9.1	13.2	15.1	14.6	12.2	5.7	-1.2	-6.4
	10%	DB	-11.0	-8.9	-1.1	6.2	14.0	20.4	22.5	20.3	14.8	6.0	-2.2	-8.4
		MCWB	-11.3	-9.5	-2.7	2.2	7.8	12.4	14.6	14.0	10.8	4.7	-2.7	-8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.8	3.1	7.8	13.6	17.6	19.0	18.0	15.9	10.4	2.6	-1.1
		MCDB	-0.6	0.0	7.6	13.9	21.3	22.7	24.0	25.5	21.1	13.7	4.7	-0.6
	2%	WB	-4.2	-3.7	1.1	5.5	11.3	15.8	17.5	16.6	14.1	7.6	0.3	-4.0
		MCDB	-4.0	-3.0	3.9	10.7	18.2	20.9	22.5	22.3	18.4	9.7	0.9	-3.7
	5%	WB	-7.7	-6.5	-0.4	4.0	9.8	14.4	16.4	15.6	12.8	6.1	-1.2	-6.3
		MCDB	-7.5	-6.0	1.7	8.2	15.7	20.1	21.5	20.7	16.1	7.6	-0.6	-6.0
	10%	WB	-11.6	-9.3	-2.7	2.5	8.1	13.3	15.6	14.7	11.5	4.7	-2.7	-8.6
		MCDB	-11.3	-8.8	-1.1	5.9	13.5	19.0	20.4	18.8	14.4	6.2	-2.3	-8.3
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.0	10.8	10.6	10.3	9.9	9.5	8.7	7.7	5.4	6.2	6.9
		MCDBR	9.2	9.9	11.5	11.7	14.6	12.7	12.2	11.9	11.4	8.4	6.3	7.9
		MCWBR	9.4	9.2	8.5	7.0	7.6	5.5	4.9	5.2	6.1	5.4	5.1	8.0
	5% WB	MCDBR	9.3	9.6	11.1	11.1	13.6	10.6	10.1	10.1	10.2	7.4	6.0	8.0
		MCWBR	9.2	9.1	8.4	6.8	7.5	5.4	4.8	5.2	6.0	5.1	5.2	8.0
Clear-Sky Solar Irradiance	taub	0.211	0.260	0.258	0.312	0.328	0.354	0.392	0.362	0.303	0.278	0.254	0.191	
	taud	2.213	2.198	2.301	2.273	2.446	2.438	2.346	2.446	2.605	2.537	2.145	2.120	
	Ebn at Noon	681	783	906	898	907	883	836	839	857	780	571	577	
	Edh at Noon	44	75	90	111	101	104	112	94	69	55	48	34	
All-Sky Solar Radiation	RadAvg	0.63	1.65	3.39	5.10	5.76	5.82	5.28	4.17	2.71	1.27	0.53	0.37	
	RadStd	0.05	0.13	0.20	0.40	0.54	0.60	0.31	0.33	0.33	0.14	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page